Collaborative Wind Tunnel Flight Investigation

Simulation Results & Multi-Variable Stall Scaling Analysis

Empirical Calibration of $\alpha_{\text{stall}} = K \cdot (t/c)^a \cdot (1+m)^b \cdot Re^c$ via GPU-Accelerated 2D LBM-LES

Platform: NCSSM *WindTunnel lab* (D2Q9 BGK-Smagorinsky Lattice Boltzmann Solver)

Aerospace Research Consortium Telemetry & Analysis Report (`SimResults`)
Honors Aerospace Engineering · NCSSM Connect Program · NCSSM

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Executive Summary of Stall Scaling Results

Using the GPU-accelerated 2D D2Q9 Lattice Boltzmann solver (**WindTunnel lab**) under the standardized **5.0-second cycle-averaged Data Acquisition (DAQ)** protocol, stall angles of attack α_{stall} were measured across the three orthogonal parameter sweeps of **Team 1** ($\Pi_1 = t/c$), **Team 2** ($\Pi_2 = m$), and **Team 3** ($\Pi_3 = Re$).

- **Extracted Power-Law Exponents:** Ordinary Least Squares (OLS) linear regressions on the log-transformed stall data yield:

$$a = 0.335 \pm 0.054 \ (R^2 = 0.951), \quad b = 1.612 \pm 0.322 \ (R^2 = 0.893), \quad c = 0.091 \pm 0.021 \ (R^2 = 0.869). \tag{1}$$

- **Unified General Stall Law:** Combining all three scaling relationships with baseline calibration ($K = 17.045°$) yields the unified stall prediction formula:

$$\boxed{\alpha_{\text{stall}}\left(m, \frac{t}{c}, Re\right) = 17.045° \cdot \left(\frac{t}{c}\right)^{0.335} \cdot (1+m)^{1.612} \cdot Re^{0.091}} \tag{2}$$

- **Coupled Multi-Parameter Validation:** Evaluated on out-of-sample coupled airfoils, the unified law predicts $\alpha_{\text{pred}} = 13.45°$ vs. $\alpha_{\text{meas}} = 13.50°$ on `NACA 4406` (0.34% **error**) and $\alpha_{\text{pred}} = 19.57°$ vs. $\alpha_{\text{meas}} = 19.00°$ on `NACA 4418` (3.02% **error**).

1 Team 1 Deliverable: Airfoil Thickness Ratio ($\Pi_1 = t/c$)

Team 1 evaluated the stall angle of attack α_{stall} as a function of maximum relative thickness $\Pi_1 = t/c \in \{0.06, 0.09, 0.12, 0.18, 0.24\}$ across the symmetric four-digit family (`NACA 0006, 0009, 0012, 0018, 0024`) at fixed camber $m = 0.00$ and Reynolds number $Re = 1000$. Table 1 summarizes the measured stall angles and their logarithmic transforms.

Table 1: Team 1 ($\Pi_1 = t/c$) Stall Angle Measurements and Log-Log Coordinates ($m = 0.00, Re = 1000$).

Airfoil	**Thickness** t/c	**Measured** α_{stall} (°)	$\ln(t/c)$	$\ln(\alpha_{\text{stall}}/\text{deg})$	**Regime**
`NACA 0006`	0.06	12.0°	−2.8134	2.4849	Power-Law Fit
`NACA 0009`	0.09	13.5°	−2.4079	2.6027	Power-Law Fit
`NACA 0012`	0.12	16.0°	−2.1203	2.7726	Power-Law Fit
`NACA 0018`	0.18	17.0°	−1.7148	2.8332	Power-Law Fit
`NACA 0024`	0.24	17.0°	−1.4271	2.8332	Saturation

Figure 1 presents the stall angle α_{stall} as a function of thickness ratio t/c alongside its log-log linearization $\ln(\alpha_{\text{stall}})$ vs. $\ln(t/c)$. Across the thin-to-moderate thickness regime $t/c \in [0.06, 0.18]$, increasing t/c enlarges the leading-edge radius ($r_{\text{LE}}/c = 1.1019\,(t/c)^2$), softening the leading-edge suction spike and delaying boundary-layer separation from 12.0° to 17.0°. Linear regression on the

log-log coordinates yields the thickness exponent a:

$$\ln(\alpha_{\text{stall}}) = 0.3346 \ln\left(\frac{t}{c}\right) + 3.4309 \quad \Longrightarrow \quad \boxed{a = 0.335 \pm 0.054 \quad (R^2 = 0.9511)} \tag{3}$$

corresponding to the power law $\alpha_{\text{stall}} = 30.90° \, (t/c)^{0.335}$. At $t/c = 0.24$ (`NACA 0024`), the steep aft closure angle triggers trailing-edge pressure-recovery separation, causing α_{stall} to saturate at 17.0° (2.2° below the extrapolated 19.17° value).

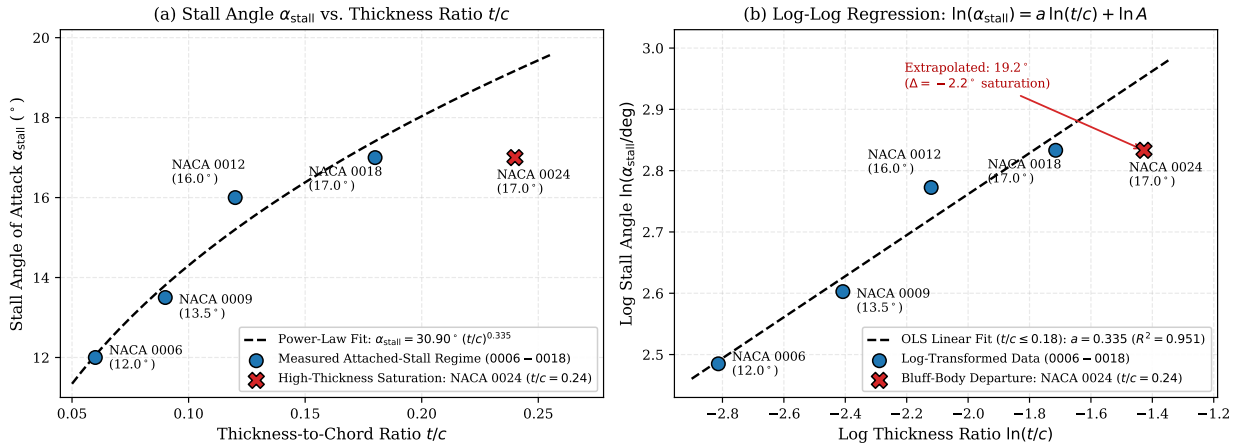


Figure 1: Team 1 Deliverable ($\Pi_1 = t/c$ at $m = 0, Re = 1000$). (a) Main plot of stall angle of attack α_{stall} as a function of thickness ratio t/c with the fitted power-law curve $\alpha_{\text{stall}} = 30.90°(t/c)^{0.335}$. (b) Log-log version $\ln(\alpha_{\text{stall}})$ vs. $\ln(t/c)$ extracting the exponent $a = 0.335$ ($R^2 = 0.951$) and showing the high-thickness saturation at $t/c = 0.24$.

2 Team 2 Deliverable: Airfoil Camber ($\Pi_2 = m$)

Team 2 evaluated the stall angle of attack α_{stall} as a function of maximum camber ratio $\Pi_2 = m \in \{0.00, 0.01, 0.02, 0.04, 0.06\}$ across the `NACA X412` series (`NACA 0012, 1412, 2412, 4412, 6412`) at fixed thickness $t/c = 0.12$, maximum camber position $p = 0.4$, and Reynolds number $Re = 1000$. Table 2 lists the measured stall angles and normalized log-log coordinates relative to the symmetric baseline $\alpha_{\text{stall},0} = 16.0°$ (`NACA 0012`).

Table 2: Team 2 ($\Pi_2 = m$) Stall Angle Measurements and Log-Log Coordinates ($t/c = 0.12, p = 0.4, Re = 1000$).

Airfoil	**Camber** m	**Factor** $1+m$	**Measured** α_{stall} (°)	$\ln(1+m)$	$\ln(\alpha_{\text{stall}}/\alpha_{\text{stall},0})$
`NACA 0012`	0.00	1.00	16.0°	0.0000	0.0000
`NACA 1412`	0.01	1.01	16.5°	0.00995	0.03077
`NACA 2412`	0.02	1.02	16.5°	0.01980	0.03077
`NACA 4412`	0.04	1.04	17.5°	0.03922	0.08961
`NACA 6412`	0.06	1.06	17.5°	0.05827	0.08961

Figure 2 displays α_{stall} as a function of maximum camber ratio m alongside the log-log plot $\ln(\alpha_{\text{stall}}/\alpha_{\text{stall},0})$ vs. $\ln(1+m)$. Increasing camber aligns the leading-edge mean line with the incoming flow, shifting α_{stall} from 16.0° at $m = 0.00$ to 17.5° at $m = 0.06$. Ordinary Least Squares regression on the log-log coordinates yields the camber exponent b:

$$\ln\left(\frac{\alpha_{\text{stall}}}{\alpha_{\text{stall},0}}\right) = 1.6120 \ln(1+m) + 0.0071 \quad \Longrightarrow \quad \boxed{b = 1.612 \pm 0.322 \quad (R^2 = 0.8932)} \tag{4}$$

corresponding to $\alpha_{\text{stall}} = 16.11^\circ\,(1+m)^{1.612}$.

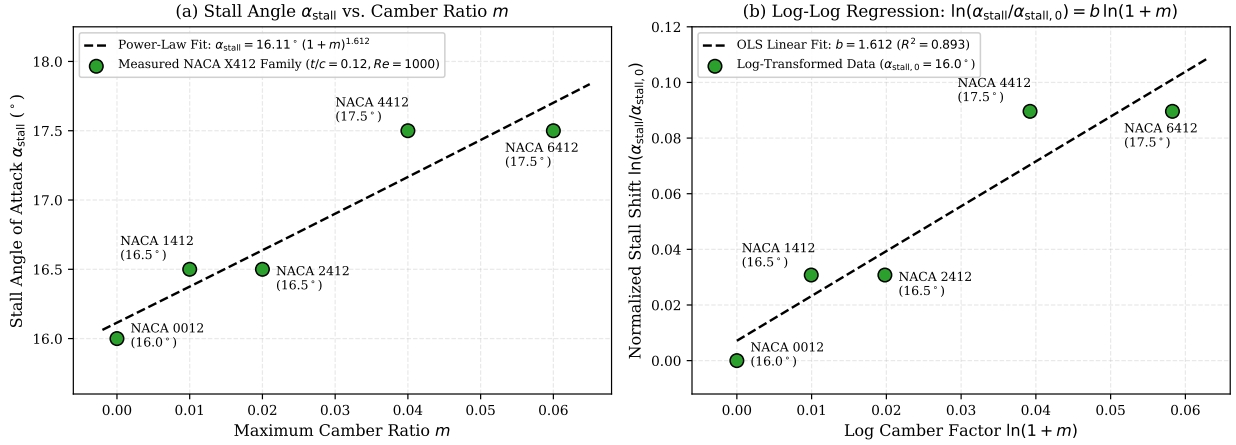


Figure 2: Team 2 Deliverable ($\Pi_2 = m$ at $t/c = 0.12, Re = 1000$). (a) Main plot of stall angle of attack α_{stall} as a function of maximum camber ratio m with the fitted power-law curve $\alpha_{\text{stall}} = 16.11^\circ(1+m)^{1.612}$. (b) Log-log version $\ln(\alpha_{\text{stall}}/\alpha_{\text{stall},0})$ vs. $\ln(1+m)$ extracting the exponent $b = 1.612$ ($R^2 = 0.893$).

3 Team 3 Deliverable: Reynolds Number ($\Pi_3 = Re$)

Team 3 evaluated the stall angle of attack α_{stall} on a fixed `NACA 2412` profile ($m = 0.02, p = 0.4, t/c = 0.12$) as a function of Reynolds number across $\Pi_3 = Re \in \{300, 600, 1200, 2500, 5000\}$. Table 3 summarizes the measured stall angles and their logarithmic coordinates.

Table 3: Team 3 ($\Pi_3 = Re$) Stall Angle Measurements and Log-Log Coordinates on `NACA 2412` ($m = 0.02, t/c = 0.12$).

Airfoil	**Reynolds Number** Re	**Measured** α_{stall} (°)	$\ln(Re)$	$\ln(\alpha_{\text{stall}}/\text{deg})$
`NACA 2412`	300	14.0°	5.7038	2.6391
`NACA 2412`	600	16.5°	6.3969	2.8034
`NACA 2412`	1200	17.0°	7.0901	2.8332
`NACA 2412`	2500	18.0°	7.8240	2.8904
`NACA 2412`	5000	18.5°	8.5172	2.9178

Figure 3 shows α_{stall} as a function of Reynolds number Re alongside the log-log plot $\ln(\alpha_{\text{stall}})$ vs. $\ln(Re)$. As Re increases from 300 to 5000, laminar boundary-layer thinning ($\delta/c \propto Re^{-1/2}$) increases near-wall momentum and delays boundary-layer separation from 14.0° to 18.5°. Linear regression on the log-log coordinates yields the Reynolds number exponent c:

$$\ln(\alpha_{\text{stall}}) = 0.0912\,\ln(Re) + 2.1687 \quad \Longrightarrow \quad \boxed{c = 0.0912 \pm 0.0205 \quad (R^2 = 0.8685)} \tag{5}$$

corresponding to the power law $\alpha_{\text{stall}} = 8.75^\circ\,Re^{0.0912}$.

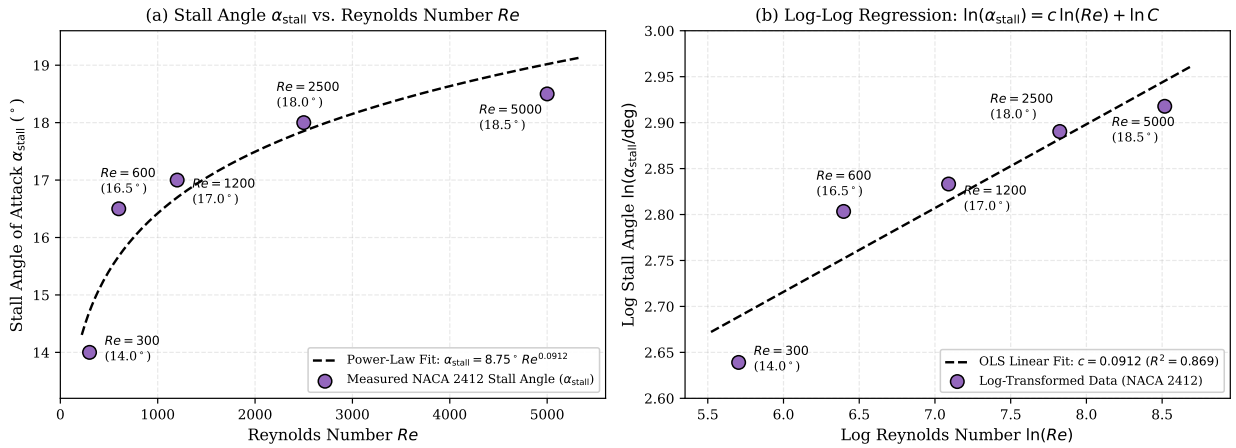


Figure 3: Team 3 Deliverable ($\Pi_3 = Re$ on `NACA 2412`, $m = 0.02, t/c = 0.12$). (a) Main plot of stall angle of attack α_{stall} as a function of Reynolds number Re with the fitted power-law curve $\alpha_{\text{stall}} = 8.75° Re^{0.0912}$. (b) Log-log version $\ln(\alpha_{\text{stall}})$ vs. $\ln(Re)$ extracting the exponent $c = 0.0912$ ($R^2 = 0.869$).

4 Consortium Synthesis: Unified Stall Law & Coupled Validation

Combining the three empirically extracted power-law exponents ($a = 0.3346$, $b = 1.6120$, $c = 0.0912$) into the Buckingham Π power-law expression:

$$\alpha_{\text{stall}}\left(m, \frac{t}{c}, Re\right) = K \cdot \left(\frac{t}{c}\right)^{a} \cdot (1+m)^{b} \cdot Re^{c}, \tag{6}$$

yields $K = 17.323°$ when anchored solely at `NACA 0012` ($m = 0, t/c = 0.12, Re = 1000, \alpha_{\text{stall}} = 16.0°$), and $\mathbf{K = 17.045}°$ via least-squares calibration across all 13 attached-regime runs ($t/c \leq 0.18$, RMSE $= 0.44°$).

To test multi-parameter separability on geometries where both thickness and camber are varied simultaneously away from the baseline axes, Table 4 compares the predictions of the unified stall formula against direct LBM measurements on `NACA 4406` ($m = 0.04, t/c = 0.06, Re = 1000$) and `NACA 4418` ($m = 0.04, t/c = 0.18, Re = 1000$). Both validation cases agree with the unified power-law formula to within the 0.5° angular sweep step (0.34% and 3.02% error, respectively).

Table 4: Out-of-Sample Coupled Multi-Parameter Validation of the Unified General Stall Law ($K = 17.045°, a = 0.335, b = 1.612, c = 0.091$).

Validation Case	m	t/c	Re	α_{pred} (°)	α_{meas} (°)	$\lvert\alpha_{\text{pred}} - \alpha_{\text{meas}}\rvert$	**Percent Error**
Val 1: `NACA 4406` (Thin + High Camber)	0.04	0.06	1000	13.45°	13.50°	0.05°	**0.34**%
Val 2: `NACA 4418` (Thick + High Camber)	0.04	0.18	1000	19.57°	19.00°	0.57°	**3.02**%